

What not to do to

24 December 2025 19:13

Weight Initialization

① ~~$w \leftarrow 0$~~

$w \leftarrow \text{small } z \leftarrow 0$

① ~~$w \leftarrow \text{same size}$~~

$w \leftarrow \text{diff size}$

① ~~$w \leftarrow \text{Small diff size}$~~

② ~~$w \leftarrow \text{Big diff size}$~~

We don't do these

③ We are assigning $w \leftarrow 0$? $\times$ (We cannot do this)

$\Rightarrow \tanh \text{ value}$

$\tanh(z) = \begin{cases} -1 \\ +1 \end{cases}$

$\tanh'(z) = 1 - \tanh^2(z)$

$\text{value}(z) = \max(0, z)$

$\text{value}'(z) = \begin{cases} 0 & \text{if } z \leq 0 \\ 1 & \text{if } z > 0 \end{cases}$

$z_{21} = a_{11}w_1 + a_{12}w_2 + a_{13}w_3 + b_{21}$

$z_{22} = a_{11}w_4 + a_{12}w_5 + a_{13}w_6 + b_{22}$

$z_{22} = 0 \rightarrow a_{22} = \tanh(z_{22})$

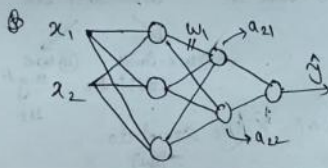
$\frac{dL}{dw_1} = \frac{dL}{dg} \frac{dg}{da_{21}} \frac{da_{21}}{dz_{21}} \frac{dz_{21}}{da_{11}} \frac{da_{11}}{dz_{11}} \frac{dz_{11}}{dw_1}$

$\frac{dL}{dw_1} = 0$

$w_1' = w_1 - \eta \frac{dL}{dw_1} \Rightarrow w_1' = w_1$

$$\Rightarrow \text{sigmoid} \begin{cases} 0 \\ 1 \end{cases} = (0, 1)$$

$$\sigma'(z) = \frac{\sigma(z)(1-\sigma(z))}{(0, 0.25)}$$



$$z_{21} = w_1 a_{11} + w_2 a_{12} + w_3 a_{13} + b_{21}$$

$$z_{21} = 0$$

$$a_{21} = \sigma(z_{21}) = \sigma(0) = 0$$

$$\frac{dL}{dw_1} = \frac{dL}{dy} \cdot \frac{dy}{da_{21}} \cdot \frac{da_{21}}{dz_{21}} \cdot \frac{dz_{21}}{da_{11}} \cdot \frac{da_{11}}{dz_{11}} \cdot \frac{dz_{11}}{dw_1}$$

$$0.25$$

$$\frac{dL}{dw_2} = \frac{dL}{dy} \cdot \frac{dy}{da_{22}} \cdot \frac{da_{22}}{dz_{22}} \cdot \frac{dz_{22}}{da_{12}} \cdot \frac{da_{12}}{dz_{12}} \cdot \frac{dz_{12}}{dw_2}$$

$$0.25$$

$$z_{22} = 0$$

$$a_{22} = 0$$

$$z_{21} = z_{22} = 0$$

$$a_{21} = a_{22} = 0$$

$$\frac{dL}{dw_1} = \frac{dL}{dw_2}$$

$$\frac{dL}{dw_1} = \frac{dL}{dw_2} = \dots = \frac{dL}{dw_n}$$

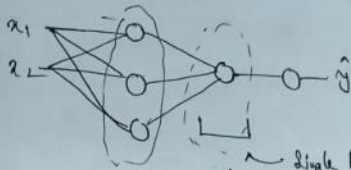
$$\text{Initially, } w_1 = w_2 = w_n = 0$$

$$w_1' = w_1 - \eta \frac{dL}{dw_1}$$

$$w_2' = w_2 - \eta \frac{dL}{dw_2}$$

$$w_n' = w_n - \eta \frac{dL}{dw_n}$$

$$\left. \begin{matrix} \text{df for upd} \\ w_1 = w_2 = w_n \end{matrix} \right\}$$



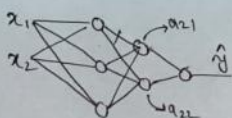
Single Perceptron
No Non linearity capture

Underfit

Q We cannot assign $w \leftarrow \text{same } N_2$?

$$\text{relu} = \max(0, z)$$

$$w = 0.5 \quad b = 0.5$$



$$z_{21} = w_1 a_{11} + w_2 a_{12} + w_3 a_{13} + b_{21} \neq 0$$

$$a_{21} = \text{relu}(0, z_{21})$$

$$z_{22} = w_4 a_{11} + w_5 a_{12} + w_6 a_{13} + b_{22} \neq 0$$

$$a_{22} = \text{relu}(0, z_{22})$$

$$z_{21} = z_{22} \Rightarrow a_{21} = a_{22}$$

$$\frac{dL}{dw_1} = \frac{dL}{dy} \cdot \frac{dy}{da_{21}} \cdot \frac{da_{21}}{dz_{21}} \cdot \frac{dz_{21}}{da_{11}} \cdot \frac{da_{11}}{dz_{11}} \cdot \frac{dz_{11}}{dw_1}$$

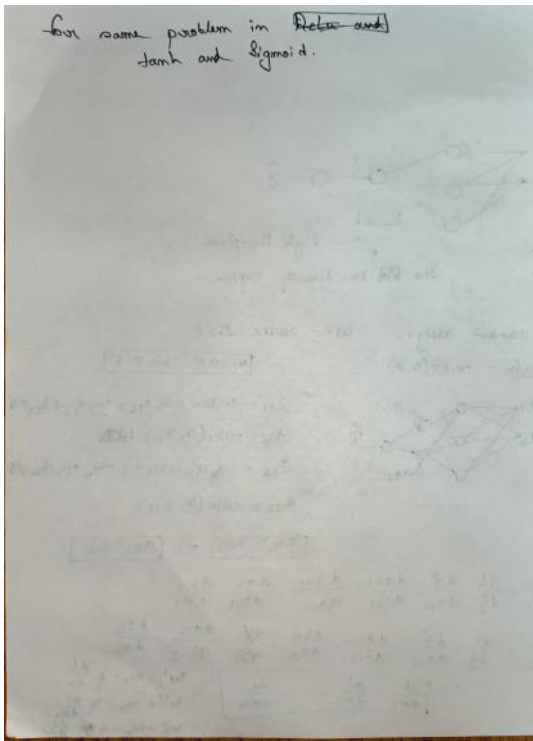
$$\frac{dL}{dw_2} = \frac{dL}{dy} \cdot \frac{dy}{da_{22}} \cdot \frac{da_{22}}{dz_{22}} \cdot \frac{dz_{22}}{da_{12}} \cdot \frac{da_{12}}{dz_{12}} \cdot \frac{dz_{12}}{dw_2}$$

$$\frac{dL}{dw_1} = \frac{dL}{dw_2} = \dots = \frac{dL}{dw_n}$$

$$w_1' = w_1 - \eta \frac{dL}{dw_1}$$

$$w_2' = w_2 - \eta \frac{dL}{dw_2}$$

$$w_n' = w_n - \eta \frac{dL}{dw_n}$$



Q Why we cannot initialize $w \leftarrow$ diff small values!

tanh

small

$$z_{11} = x_1 \underline{w_{11}'} + x_2 \underline{w_{21}'} + b_{11}$$

Small values

$$a_{11} = \tanh(z_{11}) = \frac{e^{z_{11}} - e^{-z_{11}}}{e^{z_{11}} + e^{-z_{11}}}$$

small

a_{11} is required to
update w_{11}' & w_{21}'

$$\tanh 0 = 0 \quad \tanh 0.5 = \text{und}$$

$$\tanh 45 = 1$$

$$\frac{dL}{dw_{11}'} = \frac{dL}{dy} \cdot \frac{da_{11}}{dz_{11}} \cdot \frac{dz_{11}}{dw_{11}'}$$

→ vanishing gradient.

Sigmoid

→ vanishing Gradient

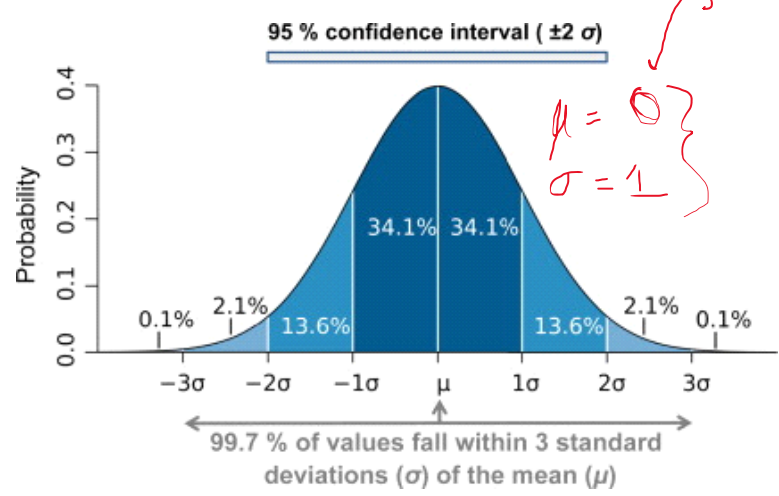
$$\sigma'(z) = (0, 0.25)$$

$$\sigma'(z) = \sigma(z) \times (1 - \sigma(z))$$

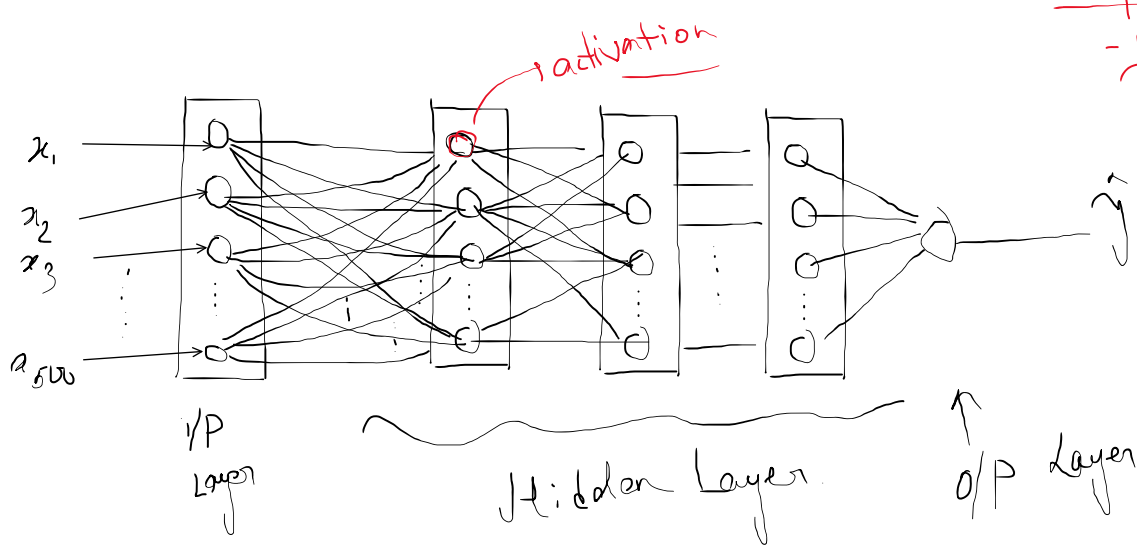
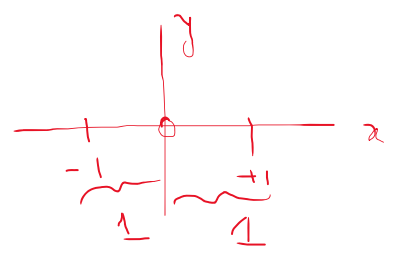
... normal distribution

$$x(1 - \sigma(x))$$

Normal distribution



x_1	x_2	...	x_{500}	x
...	...	...	...	...



$$x_i \in (-1, +1)$$

$\leftarrow$ small

$$w_{ij}^k \leftarrow \text{small}$$

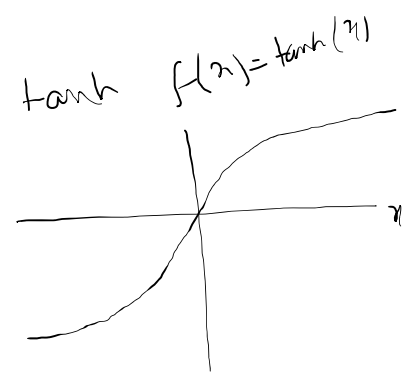
$$z_{ij} = \sum w_{ij}^k x_i \leftarrow \text{small}$$

$\rightarrow$ close to zero

$$0.007$$

$$a_{ij} = \tanh(z_{ij})$$

$\rightarrow$ close to zero

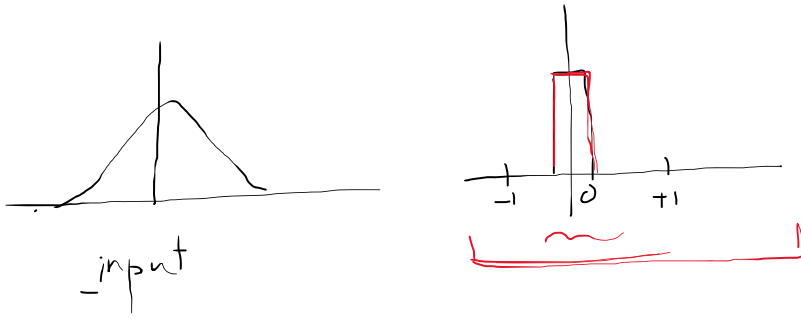


$$w_{ij}^k = w_{ij}^k - \eta \left| \frac{dL}{dw_{ij}^k} \right|$$

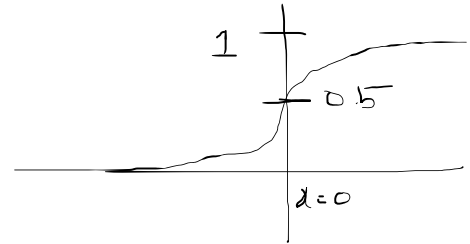
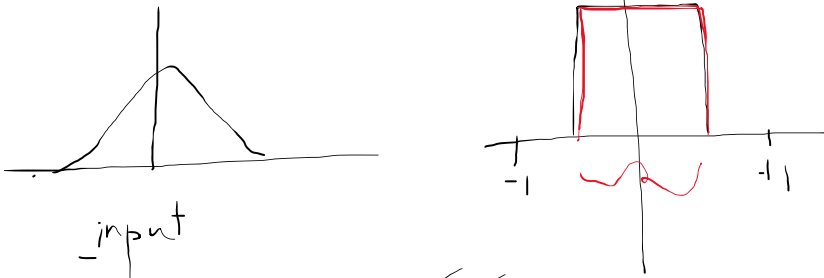
equal

$\frac{dL}{dw_{ij}^k} \leftarrow$ small

equal
VWD



Sigmoid:-



- * Deep NN ✓
- * Shallow NN ✗

relu:-
→ slow training

$\max(0, z)$
↖ close to zero
↗ close to zero

Why we cannot take large diff values.

⇒ Sigmoid & tanh:- $(-1, +1)$
↖ Saturating function

$$a = f(z)$$

Vanishing
gradient

✓
(0, 1)

$$z = \sum w_i x_i$$

↑ ↑
large large

$$a = f(z)$$

↑
Saturate

Vanishing
gradient
Descent

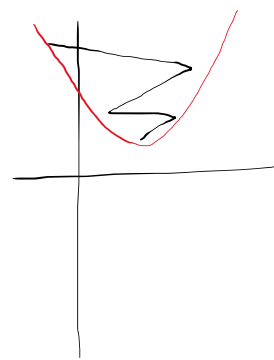
⇒ ReLU :-

→ Exploding gradient descent

$$\text{large} \rightarrow z = \sum w_i x_i$$

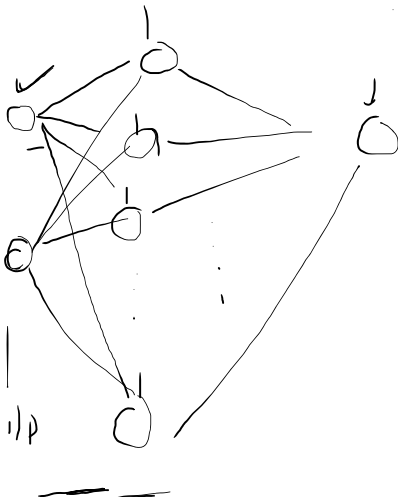
$$a_{ii} = \max(0, z)$$

↑
large



→ Unstable training
→ Late convergence

<https://www.deeplearning.ai/ai-notes/initialization/index.html>

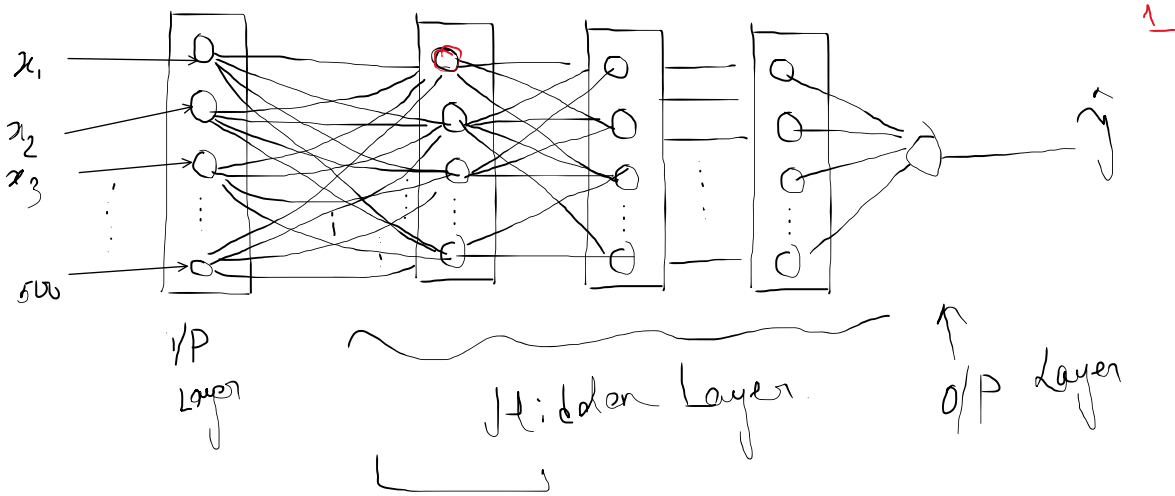


$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 1 & 2 & 3 \end{bmatrix}$$

What is small weight

$$\sqrt{\frac{2}{n_{in} + n_{out}}} * 0.01$$

np.random (n_in, n_out)



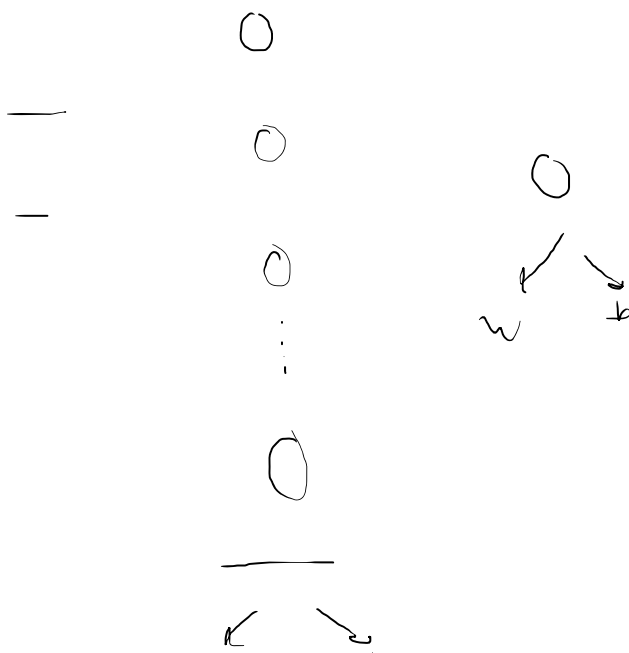
[[], [], ..., []]

shape $\rightarrow$ (n_input, n_output)

$0.01 * \leftarrow$

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[[1,1,1,1], [1,1,1], ..., [1,1,1]]

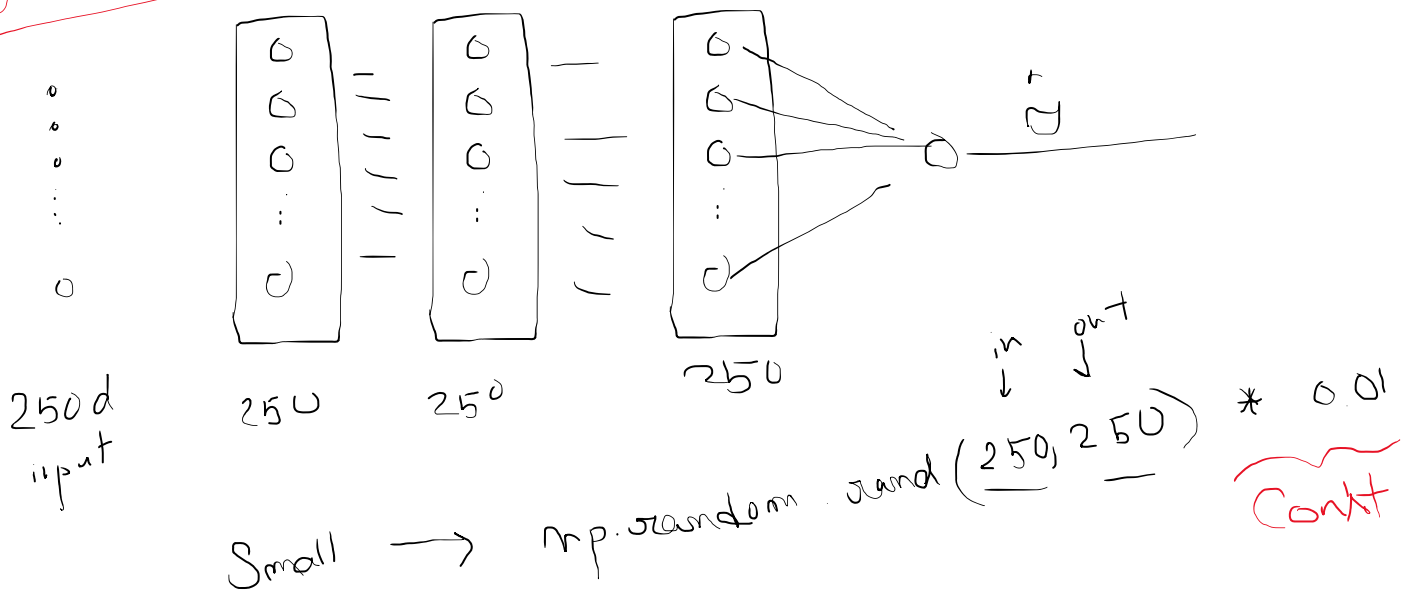


$$\begin{array}{r}
 \begin{array}{cc}
 1 & 1 \\
 w & b \\
 \swarrow & \searrow \\
 \hline
 \end{array} \\
 \hline
 (2, 10)
 \end{array}$$

→ Xavier / Glorot
(~~tanh~~, sigmoid)
Initialization technique { Normal
Uniform

→ He
(relu)
init techⁿ { Normal
Uniform

Intuition:-



Large $\rightarrow$ $\text{np.random.randn}(250, 250) * 0.8$ Const

$n \rightarrow$ # of i/p's coming to the node $\rightarrow$ $\frac{1}{\sqrt{\text{fan-in}}}$ Const

$\text{np.random.randn}(2, 2) * \frac{1}{\sqrt{2}}$

Xavier Normal = $\sqrt{\frac{1}{\text{fan-in}}}$ | He Normal = $\sqrt{\frac{2}{\text{fan-in}}}$

$$\text{Xavier Normal} = \sqrt{f_{\text{an}_{in}}} \quad | \quad \text{var} = \sqrt{\frac{2}{f_{\text{an}_{in}}}}$$

$$\text{Xavier Uniform} = [-\text{limit}, +\text{limit}]$$

$$\text{limit} = \sqrt{\frac{6}{f_{\text{an}_{in}} + f_{\text{an}_{out}}}}$$

$$\text{He Uniform} [-\text{limit}, +\text{limit}]$$

$$\text{limit} = \sqrt{\frac{6}{f_{\text{an}_{in}}}}$$